Notes: Crémer, Garicano & Prat (QJE, 2007)
Language and the Theory of the Firm

Contents

1 Big picture

- **Question.** How do specialized technical languages shape firm scope, integration, hierarchy, and decentralization?
- **Core friction.** Agents are boundedly rational and can learn only a limited number of words.
- **Code as partition.** A language is a partition of possible problems. A broader word is less precise and requires more diagnosis.
- **Optimal language result.** Efficient codes use precise words for frequent events and vague words for rare events.
- **Firm-scope trade-off.** Broad organizational scope captures synergies but requires either a common code or a translator/hierarchy.
- **Economic takeaway.** Language is an organizational asset that determines specialization, coordination, firm boundaries, and the role of managers.

2 Incremental contribution

- The paper formalizes Arrow's idea of organizational codes as a partition of events under bounded rationality.
- It connects language design to the theory of the firm, adding a communication-based theory of scope to property-rights and incentive theories.
- It shows why specialization and coordination are in tension: specialized codes are locally precise but hinder communication across units.
- It explains hierarchy as translation and shows that common codes and hierarchy are substitutes.
- It explains the IT-era combination of centralized information and decentralized decision making: common codes reduce the need for managerial translators.

3 Model primitives and measurement

A code $C = \{W_1, \ldots, W_K\}$ is a partition of the set of possible problems X. Word k has breadth n_k and frequency p_k. Bounded rationality limits the number of words agents can learn.

$$D(C; f) = \sum_{k=1}^{K} p_k d(n_k).$$

Interpretation. A word is a compression device. It saves vocabulary capacity but creates ambiguity for the receiver.

4 Models and theoretical specifications

4.1 Optimal codes

Proposition 1 says broader words describe less frequent events. Proposition 2 strengthens this when diagnosis costs are convex in word breadth. Proposition 3 says more complex environments have higher minimal diagnosis costs. Proposition 4 shows that the value of additional words depends on environment complexity.

4.2 Shared codes and no dialects

When an engineer communicates with two salesmen facing different event distributions, the baseline model rules out optimal dialects. Proposition 5 states that optimal codes contain K words and satisfy $C_A = C_B$.

Interpretation. Dialects use up scarce vocabulary capacity. A common code avoids wasting bounded rationality.

4.3 Organization: separation, integration, hierarchy

Separation preserves specialized service codes but forgoes synergies. Integration uses a common code and captures synergies but worsens local diagnosis. Hierarchy keeps separate codes and adds a translator at fixed cost μ.

$$\Pi^S = q^{NC}\left[1 - \lambda(v_A D^*(f_A) + v_B D^*(f_B))\right],$$
$$\Pi^I = q^C\left[1 - \lambda D^*(v_A f_A + v_B f_B)\right].$$

Proposition 6 says integration is favored by lower diagnosis cost, higher synergy, and more homogeneous service distributions. Proposition 7 gives thresholds for integration, hierarchy, and separation.

5 Theoretical design and empirical implications

The theory predicts that lower information/search costs increase common code adoption, horizontal links, and decentralization. ERP systems, Microsoft, and the B-2 bomber cases illustrate how common technical grammars can substitute for managerial translation.

6 Figures and extracted proposition blocks

6.1 Figure I: Communication in Three Possible Organizational Forms

Read: Separation preserves local language; integration uses a common code; hierarchy uses a translator.

Economic message. Organizational form is a communication architecture.

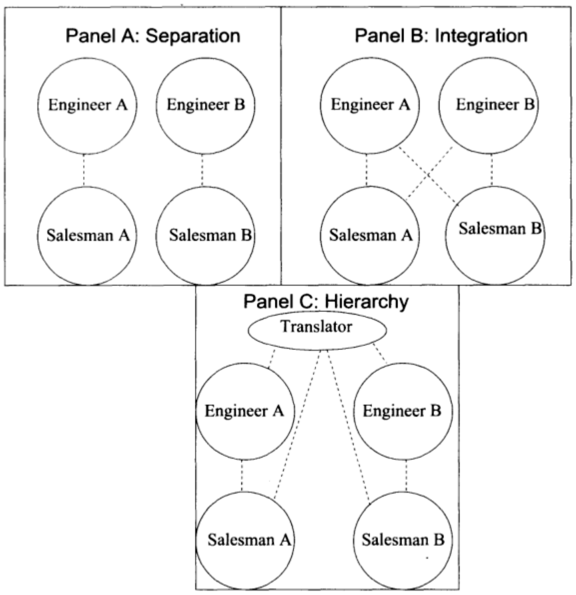

FIGURE I
Communication in Three Possible Organizational Forms
The dashed lines represent lines of communication.

(a) Figure I: three organizational forms

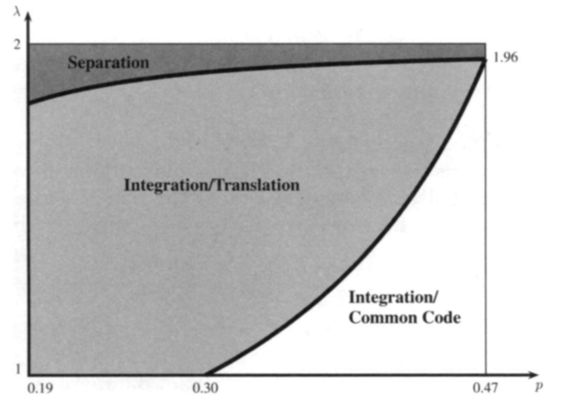

FIGURE II
The Choice Between Separation With or Without a Hierarchy for Exampl
s a function of the cost of communication λ and of the synergy paramete

(b) Figure II: optimal form by diagnosis cost and synergy

6.2 Model and profit blocks

The bounded rationality of the agents is represented b
maximum number of words, K, that they can learn.

Having received word k from the salesman, the eng
must still identify the precise problem x of the client in or
solve it. This *diagnosis* stage takes time and/or energy, a
cost is a strictly increasing function d of the breadth n_k of
k—the less precise the word, the more work diagnosis req
The expected diagnosis cost associated with code C is there

$$D(C; f) = \sum_{k=1}^{K} p_k d(n_k), \tag{1}$$

(a) Diagnosis cost block

the salesman with no client. Then,

$$q^{\mathrm{C}} = 2(1 - p + p^2),$$

with the clients served by the other engineer are determined by

$$\phi = \frac{1 - p}{1 - p + p^2}.$$

The aggregate profits of the two services are given by th
value of the clients serviced minus the per client diagnosis cos
We let $\lambda D^*(f)$ be the output cost of diagnosis associated with
distribution of events f when the optimal code is chosen, that i

$$\lambda D^*(f) = \min_C \lambda D(C; f).$$

where λ, the "diagnosis cost" parameter expresses diagnosis co
in output units.[13]

Then we have that the profits of the two services whe
nonintegrated are given by

$$\Pi^S = q^{\mathrm{NC}}[1 - \lambda(v_A D^*(f_A) + v_B D^*(f_B))]. \tag{4}$$

When services do collaborate, salesmen need to be able
communicate customer needs to both engineers. The proof
Proposition 5 can easily be adapted to show that the two servic
must use the same code, and the proof of Corollary 1 can l

(b) Profit under separation and integration

Economic message. The equations translate language precision into firm productivity.

6.3 Propositions 1–5: efficient languages and common codes

II.B. Optimal Codes

In this subsection, we derive some properties of the opti code for these two agents (remember that all proofs are in Appendix).

PROPOSITION 1. In an optimal code, broader words describe frequent events: if $n_k > n_{k'}$, then $f_x \leq f_{x'}$ for any $x \in W_k$

(a) Proposition 1

is (weakly) convex.

PROPOSITION 2. If the function d is "convex" in the number events n, i.e., if

$$d(n+1) - d(n) \geq d(n'+1) - d(n') \quad \text{for all } n \geq n' \geq 1,$$

then, unless integer constraints make it impossible, in optimal code broader words are used less frequently: if n_k $n_{k'} \geq 2$, then $p_{k'} \geq p_k$.

(b) Proposition 2

bution f. Similarly, let $\bar{F}_i$ denote the probability that the event ha index i or lower given distribution $\bar{f}$ (but recall that we are stil using the indexing derived using f). We then say:[10]

DEFINITION 1. The distribution f represents a more complex en vironment than $\bar{f}$ if $F_i \geq \bar{F}_i$ for all events i.

An environment becomes simpler if unlikely events becom even less likely and likely events become even more likely. In th three-event example above, for any $\bar{p} > p > 1/3$, the distributio with p represents a more complex environment than the distri bution with $\bar{p}$.

We can show that the diagnosis cost is increasing in complexity

PROPOSITION 3. If f represents a more complex environment tha $\bar{f}$, the minimal diagnosis cost associated with f is (weakly

(a) Definition 1 and Proposition 3

implies an interesting interaction between the benefit of addi tional words and the complexity of the environment.

PROPOSITION 4. Increasing the number of words from 1 to $K >$ lowers diagnosis costs more for less complex environments On the other hand, moving from K words to a very larg number of words (perfect communication) lowers diagnosi more for more complex environments.

(b) Proposition 4

and B knows 2.

Proposition 5 shows that the same code, which saturates t rationality constraints of all the agents, will be used in comm nicating with both salesmen.

PROPOSITION 5. The optimal codes contain K words and satis

(a) Proposition 5

distributions of problems in the two distributions of clients ar We define the homogeneity of the two client distributions fo mally next.[14]

DEFINITION 2. Given a client population f, the client distributio among services $(\bar{f}_A, \bar{f}_B, \bar{v}_A, \bar{v}_B)$ is more ***homogeneous*** than th client distribution (f_A, f_B, v_A, v_B) if the distribution over th entire population f is the same under both and the distribu tions of types $\bar{f}_A$ and $\bar{f}_B$ are convex combinations of f_A and f_B.

III.B. Integration or Separation?

In this subsection, we study the choice between integrate and separated services focusing on the comparative statics of th choice. Intuitively, it is clear that the key loss due to collaboratio results from the deterioration in within-service communicatio when a less specialized language is used. We show this next.

The fundamental result on which our analysis rests is th concavity of the diagnosis cost.

LEMMA 1. For any two distributions f_A and f_B and any $\alpha \in [0, 1$ we have

$$D^*(\alpha f_A + (1-\alpha) f_B) \geq \alpha D^*(f_A) + (1-\alpha) D^*(f_B).$$

(b) Definition 2 and Lemma 1

Economic message. Frequent problems receive precise words; dialect variety is costly because it consumes scarce vocabulary capacity.

6.4 Propositions 6–7: scope, hierarchy, and thresholds

parative statics characterizing the choice of organizational form

PROPOSITION 6. An integrated form becomes relatively more prof itable compared to the segregated form as (a) the diagnosi cost λ decreases (b) the synergy between the two services measured by q^C/q^{NC}, increases, or (c) the homogeneity of th two client distributions increases.

Parts (a) and (b) are easily proven by examination of (4) an

(a) Proposition 6

PROPOSITION 7. There exists $\mu^* > 0$ such that, for all $\mu \in (0, \mu^*$ there exists $\lambda^{IH}(\mu)$ and $\lambda^{HS}(\mu)$ such that the unique optima organization is

Integrated	if $\lambda < \lambda^{IH}(\mu)$,
Hierarchical	if $\lambda \in (\lambda^{IH}(\mu), \lambda^{HS}(\mu))$,
Separated	if $\lambda > \lambda^{HS}(\mu)$.

The function λ^{IH} is decreasing, while λ^{HS} is increasing.

(b) Proposition 7

serve $v_A \phi q^C$ clients of salesman A and $v_B(1-\phi)q^C$ clients of sales man B. Hence, the language that the translator will use to speal to him will be the language appropriate for the distribution

$$\frac{v_A\phi}{v_A\phi + v_B(1-\phi)} f_A + \frac{v_B(1-\phi)}{v_A\phi + v_B(1-\phi)} f_B.$$

The profits from the hierarchical organization will therefore b

$$(6) \quad \Pi^H = q^C\Bigg[1 - \lambda\Bigg[(v_A\phi + v_B(1-\phi))D^*\left(\frac{v_A\phi f_A + v_B(1-\phi)f_B}{v_A\phi + v_B(1-\phi)}\right) + (v_A(1-\phi) + v_B\phi)D^*\left(\frac{v_A(1-\phi)f_A + v_B\phi f_B}{v_A(1-\phi) + v_B\phi}\right)\Bigg]\Bigg] - \mu$$

(a) Hierarchy profit formula

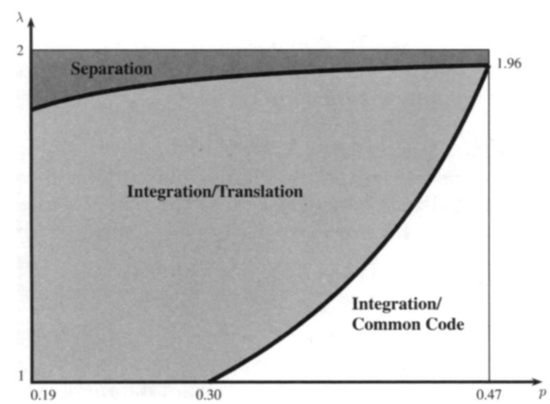

FIGURE II

The Choice Between Separation With or Without a Hierarchy for Exampl s a function of the cost of communication λ and of the synergy paramete

(b) Figure II revisited

Economic message. Integration, separation, and hierarchy are chosen according to diagnosis costs, synergies, and translation costs.

7 Short summary

- Technical language is modeled as a code: a partition of possible problems.
- Bounded rationality limits vocabulary size.
- Optimal codes assign precise words to frequent events and vague words to rare events.
- Common codes enable coordination but reduce local specialization.
- Hierarchy is a translator between specialized codes.
- IT lowers diagnosis costs and substitutes common codes for hierarchical translation.

8 Conclusion

8.1 Core conclusion

Language is a central determinant of organizational form. The scope and hierarchy of firms depend on whether agents can share a code or need translation.

8.2 Incremental contribution revisited

The paper's key contribution is to make organizational language a formal economic object and to connect it to scope, integration, hierarchy, and delayering.

8.3 Economic intuition

A word compresses information. Compression saves bounded rationality but creates ambiguity. Organizations trade local precision against broad coordination.

8.4 Implications

Common databases and standardized information systems can centralize information while decentralizing decisions because they substitute for translators.

8.5 Limitations

The model abstracts from incentives, authority, and strategic communication, and the empirical evidence is mainly illustrative.

8.6 Takeaway

A firm is not only a nexus of contracts or assets; it is also a nexus of shared meanings.